

## Examiner reports

### Q1.

This question was very well answered.

### Q2.

The quality of solutions to problems set on this area of the specification has improved steadily since the first paper in 2001. It was pleasing to see a variety of methods being used and many candidates scored well on this demanding question.

Part (a) was attempted successfully either using the equations of constant acceleration or by integration. The most common error was an incorrect sign in the acceleration of the vertical component.

Part (b) proved to be the most demanding part of the question. Again, varied methods were used, ranging from a full verification of the result to the quotation of the general equation of the path of the trajectory with appropriate substitutions.

Part (c) was popular, yielding marks to most candidates.

The varied methods used in part (d) revealed clear understanding of the motion. The most popular methods used the result quoted in part (b) or the equations of motion with constant acceleration, sometimes incorporating the symmetry of the motion.

In part (e), for those candidates who realised that the greatest speed occurred either at  $O$  or at  $C$ , the request was straightforward and full marks were often obtained.

### Q3.

This was a straightforward question for those who were able to quote and use the equations of constant acceleration accurately. Assumptions quoted in part (b) were varied and mostly valid.

### Q4.

Part (a) was mostly completed successfully, showing a pleasing improvement in algebraic skills. Part (b) proved more discriminating. Many candidates found sketch graphs helpful

in understanding the problem. The most direct method was the use of  $s = \frac{1}{2}(u+v)t$ , but many were successful with longer methods involving the retardation.

### Q5.

This last question revealed an improved understanding of projectile motion, as well as clear and accurate arithmetic work. A variety of methods were used in all parts of the question, including use of the symmetry of the motion.

Part (a) was done well, with vertical motion being considered both upwards and downwards. In part (b), again both directions of the vertical motion were used. Part (i) was done well, but part (ii) proved more discriminating, with a weakness in solving quadratic

equations being shown by some. Part (iii) was again mostly successfully and accurately completed.

**Q6.**

Parts (a) and (b) of this question proved very successful. Some candidates had difficulty in appreciating the nature of the motion after the string broke and chose an incorrect acceleration in part (c). Graphs in part (d) were generally good.

**Q7.**

Diagrams in part (a) were good, but algebraic expressions produced in part (b) were poor. The use of the simple principle 'speed = distance / time' proved beyond many candidates. Many seemed unaware of what constitutes an 'equation', omitting the equals sign. There were, however, some excellent solutions to this question, showing clear and concise work.

**Q8.**

Again the poor use of algebra proved to be the downfall of many candidates. Part (a) was answered quite well, with a satisfactory proportion of candidates reaching the printed solution, but there were many careless errors in part (b). Part (c) was generally successful, but few were able to apply the same principle in part (d), with the given speed often being treated as a vertical velocity component.

**Q9.**

Most candidates found this question very straight forward, with very many gaining full marks. A few made minor errors. In part (a) some students simply divided 80 by 20 to give a time of 4 seconds.

In part (b), some candidates introduced the weight of the van. In the final part of the question, some candidates used the correct terms, but introduced the wrong signs.

**Q10.**

This question well answered by the candidates with a number gaining full marks. The candidates seemed to have been helped by the printed answer given in the first part.

**Q11.**

Part (a) of this question was generally done very well, although some unusual vector manipulation was observed on a few scripts. Several candidates worked with each component and combined their results into a vector, rather than working with vectors throughout their solutions.

In part (b), a large number of candidates found the velocity correctly, but did not then go on to find the speed of the particle.

There were a fair number of good attempts at part (c). Some candidates made errors with the signs when first setting up their equations. There were a few candidates who did not make any significant attempt at this part of the question.

**Q12.**

The printed answer helped candidates to do well in part (a). A few candidates did not show enough working to justify the allocation of all three marks. There were some candidates who worked with equations for the vertical component of the motion, rather than the horizontal component.

Part (b) was again well answered. A few candidates made arithmetic errors and some found the maximum height instead of the height of the crossbar.

Candidates found part (c) very difficult. Many of them considered the position of the ball rather than its velocity. Some considered only one component of the velocity. However there were a few candidates who did produce a good solution to this part of the question. Most of the candidates stated appropriate assumptions, although some candidates produced inappropriate assumptions.

### Q13.

Parts (a) to (c) of this question were answered well by many of the candidates, but part (d) caused more problems. There were candidates who simply multiplied the mass and the acceleration, ignoring the resistance force. The other error was to obtain 4600 instead of 4200.

### Q14.

Part (a) was generally answered well with a mixture of integration and constant acceleration equation methods.

Part (b) was done better than expected, with only a small number of candidates working with the wrong component.

There were a variety of minor errors in part (c), but also many correct solutions.

Many candidates found part (d) too demanding and made little progress. It was not uncommon to see solutions that still contained an acceleration or that ignored the initial position of the particle.

### Q15.

There were many good responses to the first two parts of this question and some candidates were clearly helped by the answer given in these parts. Part (c) was more challenging and a number of candidates produced some quite confused work. The main issue was that they were using incorrect heights and/or incorrect initial velocities.

### Q16.

This question was generally done well by the majority of candidates, but some experienced problems. In part (b), there were a few candidates who ignored the instruction to form an equation of motion for each particle. A few others made errors in one equation. Part (c) was usually done correctly if the candidates had two equations of motion in part (b). If they did not have these equations, there were often errors in this part. Part (d) was done well using the acceleration given in the question.

### Q17.

Part (a) was done well by many candidates, who may have been helped by the printed

answer. There were also some candidates who made little progress and did not seem to be able to see how to form an equation to find the time. Part (b) was often done well, but was completed by fewer candidates than part (a). Some candidates did not seem to be able to form a correct expression for the height. A few candidates used

$h = 10 \sin 60^\circ t + \frac{1}{2} g t^2$  instead of  $h = 10 \sin 60^\circ t - \frac{1}{2} g t^2$ . Part (c) proved to be very challenging.

Many candidates made very little progress with this part of the question. Some found the vertical component of the velocity when the ball hit the wall. Only a small proportion of the candidates could incorporate the horizontal component and find the correct speed.

### Q18.

Many candidates were able to complete part (a), probably helped by the printed answer. Many of these candidates did not produce any further correct working. In part (b), some candidates could write down correct expressions for the position vector. A common error was to apply the constant acceleration formula  $\mathbf{r} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t$  using the two velocities given in the question without realising that  $\mathbf{v}$  was variable.

Of the candidates who had a correct answer in part (b), a fair proportion went on to try part (c). There were a reasonable number of correct responses, but there were a number who used the wrong component or a mixture of the two. There were very few correct responses to part (d). The main incorrect approach was to use the position vector. Those candidates who realised they needed to find an expression for the velocity generally went on to complete this part of the question.

### Q19.

Part (a) of this question was done very well and posed little difficulty for the vast majority of candidates. Part (c) was a little more demanding. Some candidates simply found the product  $ma$  giving the answer 104 N; others included three terms, but made a sign error in their equation which typically produced a final answer of 69 N.

### Q20.

Parts (a) and (b) proved to be quite accessible for most candidates, but part (c) was very much more demanding. In part (a) the candidates tended to score all or no marks, with a complete solution following a correct first step. There were occasional arithmetic errors in the working of some candidates. A correct answer to part (a) was often followed by the velocity in part (b). A few candidates used incorrect formulae or tried to use an expression for the position vector rather than the velocity. Part (c) caused a variety of problems, with some candidates forming equations that contained vectors. A few produced an equation that would have arisen from moving south west instead of south east. A correct equation was normally followed by a complete solution. A very small number of candidates found the velocity, but not the speed of the boat.

### Q21.

This question was generally done very well and many candidates gained full marks. In part (c), a few candidates multiplied the mass of the car by 9.8 instead of 1.2.

**Q22.**

The candidates found this question a little harder than the previous two questions. There were a number of poor force diagrams, some showing a force in the direction of motion. The errors made by candidates were mainly due to incorrect resolving, but some of these were corrected to produce the answers given in the question. A small number stated that the reaction force was 49 N.

**Q23.**

The candidates who chose the appropriate constant acceleration equation in part (a) had little difficulty. However, a number did treat the position vector as if it were a velocity vector. In part (b), some candidates stopped when they had found the velocity and did not go on to find the speed as requested. A few candidates had problems dealing with the negative signs. Part (c) was very challenging. Some candidates made good attempts at this, but they were very much in the minority. There were quite a few reasonable answers to part (d).

**Q24.**

This question was generally done very well, especially parts (a) and (b). There were, in the main, only minor errors made on these parts. Part (c) did cause difficulties for a small number of candidates. The main problems were that they either omitted the weight or simply calculated it. A few did use the correct terms in an equation that contained a sign error.

**Q25.**

This question was done well by quite a few candidates. Many of these found the acceleration as part of their solution to part (a). The candidates who had difficulties were those who did not realise that they needed to use a constant acceleration formula or those who quoted an inappropriate formula as their starting point. A few candidates did have difficulty with part (b), but several who had struggled with part (a) used the printed result to enable them to attempt part (b). Those who had been successful with part (b) normally went on to complete part (c).

**Q26.**

The great majority of candidates answered this question very well. There were a few minor slips in parts (a) and (b). Part (c) did cause more difficulty for some candidates. The main error was to use 5m as the distance instead of subtracting this from the maximum height.

**Q27.**

Part (a) was usually done very well. However there were very few good responses to part (b). The most common error was to try to find the magnitude of the resultant of  $P$  and the friction, assuming that the friction force had magnitude 58.8 N. For many candidates their understanding of the friction inequality seemed poor. In part (c), the candidates tended to either find the correct value or to simply give the answer 16 N, which was the product of the mass and the acceleration. Many candidates did part (d) very well, but a few omitted the negative sign from the acceleration. Those candidates who had difficulties usually used an incorrect acceleration.

**Q28.**

In part (a), there were a number of candidates who simply seemed to think that the acceleration was given by distance over time. Many of these candidates also gave the units as  $\text{m s}^{-1}$ , which further confirmed that their approach was incorrect. However their attempts at part (b) were very much better, probably helped by the printed answer. There were also a number of good solutions to part (c). The most common error in part (c) was to ignore the acceleration when calculating the friction force.

**Q29.**

Part (a) was done very well by a good number of the candidates. A few worked with the magnitudes of the quantities, starting with the magnitude of the force, rather than with the actual vectors. Part (b) was found to be much more demanding. The two main errors made by the candidates were to assume that the particle still had the same acceleration as in part (a) or to use a time of 45 s rather than 25 s for the second stage of the motion.

**Q30.**

There were a fair number of good responses to part (a). The candidates who did not produce a correct solution generally either missed out the difference between the initial and final heights of the ball, or made sign errors in the setting up of their equations. A few candidates made arithmetic errors. Part (b) was also done well, with some candidates being helped by the printed answer in part (a). A few candidates used incorrect times from part (a) when trying to find the range. There were only a few correct solutions to part (c). An incomplete solution, which only found the vertical component of the velocity, was quite common among those who made some progress with the solution.

**Q31.**

Parts (a) and (b) of this question were done very well by almost all of the candidates. Part (c) proved to be a little more challenging. One common error was simply to calculate the product of the mass and the acceleration. The other common error was to include the correct terms but to make a sign error.

**Q32.**

This question was answered very well by almost all of the candidates, with many complete solutions. A small number of candidates did make minor mistakes, often of an arithmetic nature.

**Q33.**

This question was generally done well. Some candidates lost marks in part (a) because they ignored the instruction "By forming an equation of motion for each particle...". A few candidates produced a negative acceleration in part (a), which was satisfactory because it was consistent with their equations, but then substituted a positive acceleration into an equation in part (b) to find an incorrect value for the tension.

There were many candidates who answered part (c) well including some of those who had had difficulty with the previous parts.

**Q34.**



Parts (a) and (b) were quite well answered; particularly part (a) where the answer was given.

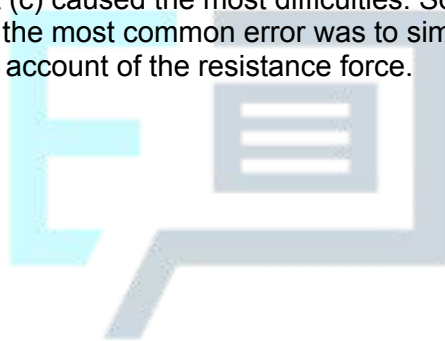
In part (b) a few candidates confused the initial and final velocities. Some candidates found the position by assuming that at time  $t$  the velocity of the aeroplane would be  $19\mathbf{i}+13\mathbf{j}$ .

Part (c) caused a great deal of difficulty for some candidates. Many did not separate the two components of the position vector and tried to find  $t$  by working with the whole vector. One consequence of this was that final answers for the time were given as vectors. Those candidates who formed separate equations for each component of the position vector were more successful, but some did make errors solving the equations.

### Q35.

The majority of candidates made a good attempt at this question and there were many completely correct solutions. In part (a), some of the candidates considered the three stages of the motion, but made errors when calculating one or more of these distances. In part (b), a very small number of candidates confused the numbers and gave the reciprocal of the acceleration as their answer. The majority of force diagrams in part (c) were correct, but the most common error was to omit one force.

Calculating the force in part (c) caused the most difficulties. Some candidates made a sign error in their equations, but the most common error was to simply multiply the acceleration by the mass, without taking account of the resistance force.



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